LOTLINE • RECALL INTELLIGENCE PLATFORM

INTERNAL INCIDENT INVESTIGATION DOSSIER

REPORT ID: LOTLINE-RPT-INC-2026-001-151743
GENERATED: 2026-11-10 15:17:43 UTC

SYNTHETIC BENCHMARK SCENARIO • DEMO MODE

SECTION 01: INCIDENT OVERVIEW & VERIFIED IDENTITY

Direct telemetry queried from IBM Db2 table LOTLINE_APP.INCIDENTS.

Incident Identifier:	`INC-2026-001`	**Internal Severity:**	**CRITICAL** (Internal Risk Assessment)
Incident Status:	**OPEN**	**Biological Pathogen:**	**Salmonella** (Supplier QA Alert)
Contaminated Source Lot:	`RAW-204`	**Raw Material Description:**	Organic Spinach
Original Raw Quantity:	**500.0 KG (Raw Mass)**	**Source Lot Status:**	**QUARANTINED**
Supplier Identity:	GreenLeaf Foods	**Supplier Country / State:**	India (No street address in Db2)

6	2,400	1 / 4	2
DOWNSTREAM BATCHES (3 WIP + 3 Finished)	FINISHED UNITS EXPOSED (100% Commercial)	APPROVED ACTIONS (Human Confirmed)	PRODUCTION LEVELS (Recursive Lineage)

PATHOGEN CHARACTERIZATION & OPERATIONAL SCOPE

Biological Threat Profile: *Salmonella enterica* is a virulent foodborne bacterial pathogen capable of causing acute salmonellosis (fever, abdominal cramps, severe diarrhea). In raw agricultural commodities such as spinach, vegetative internalization or surface adherence poses an immediate public health hazard. Because intermediate puree blending (Tier 1) operates without a verified thermal lethality step, cross-contamination propagates directly into commercial finished lots.
Regulatory Boundary Notice: This investigation represents an internal facility risk evaluation based on Db2 ground truth. Statutory hazard classifications (such as formal FSSAI Class I recall mandates) are issued exclusively by the Designated Officer.

METRIC GROUNDING DIRECTIVE (STRICT UNIT SEPARATION): Raw ingredient inventory is measured in mass (**KG**) and must never be mathematically summed with downstream commercial packaged products (measured in individual **UNITS**). Raw mass and packaged units are non-convertible metrics and are reconciled independently across IBM Db2 inventory and shipment tables.

SECTION 02: CONTAMINATION TRACEABILITY & MATERIAL MASS BALANCE

Lineage determined via IBM Db2 Recursive Common Table Expression (WITH RPL) querying LOTLINE_APP.LOT_LINEAGE. Total downstream nodes: **6** | Depth levels: **2**.

TIER 0: SOURCE RAW INGREDIENT	TIER 1: INTERMEDIATE WIP BATCHES	TIER 2: FINISHED COMMERCIAL GOODS
RAW-204 Organic Spinach Initial: 500 KG Quarantine: 50.0 KG Facility: FAC-NGP	**WIP-301** (Spinach Wrap Base) • Consumed: 180.0 KG | Facility: FAC-NGP **WIP-302** (Spinach Smoothie Puree) • Consumed: 120.0 KG | Facility: FAC-NGP **WIP-303** (Spinach Bowl Mix) • Consumed: 100.0 KG | Facility: FAC-NSK	**FIN-401** (Green Spinach Wrap) • Created: 1,000 UNIT | Expiry: 2026-11-20 **FIN-402** (Spinach Green Smoothie) • Created: 800 UNIT | Expiry: 2026-11-18 **FIN-403** (Garden Veggie Bowl) • Created: 600 UNIT | Expiry: 2026-11-18

DETERMINISTIC RAW MATERIAL MASS BALANCE RECONCILIATION

Audit reconciliation between intake mass, downstream consumption, and warehouse quarantine.

Intake Raw Lot	Original Intake	Recorded WIP Consumed	Quarantined Stock	Unaccounted Variance	Audit Status
RAW-204	**500.0 KG**	400.0 KG (WIP-301..303)	50.0 KG (FAC-NGP)	**50.0 KG**	**UNRECONCILED**

MASS RECONCILIATION AUDIT FINDING: A discrepancy of **50.0 KG** exists between the original intake lot (500.0 KG) and the sum of recorded WIP production (400.0 KG) plus remaining quarantined storage (50.0 KG). Per regulatory audit rules, this difference is explicitly designated as **UNRECONCILED**. Automated balancing or inventory write-offs without physical inspection records are strictly prohibited.

RECURSIVE TRACEABILITY CTE RECORD AUDIT

Depth	Parent Lot	Child Lot	Product Name	Lot Type	Facility	Output Quantity
Level 1	RAW-204	WIP-301	Spinach Wrap Base	**WIP**	FAC-NGP	**180 KG**
Level 1	RAW-204	WIP-302	Spinach Smoothie Puree	**WIP**	FAC-NGP	**120 KG**
Level 1	RAW-204	WIP-303	Spinach Bowl Mix	**WIP**	FAC-NSK	**100 KG**
Level 2	WIP-301	FIN-401	Green Spinach Wrap	**FINISHED**	FAC-NGP	**1,000 UNIT**
Level 2	WIP-302	FIN-402	Spinach Green Smoothie	**FINISHED**	FAC-NGP	**800 UNIT**
Level 2	WIP-303	FIN-403	Garden Veggie Bowl	**FINISHED**	FAC-NSK	**600 UNIT**

SECTION 03: COMMERCIAL EXPOSURE & LOGISTICS INTERVENTION

Commercial finished-goods inventory distribution queried from LOTLINE_APP.INVENTORY and LOTLINE_APP.SHIPMENTS. Total exposure: **2,400 finished product units**.

Batch Lot	Product Name	Warehouse (Hold)	Delivered Retail	In Transit	Total Exposure
FIN-401	**Green Spinach Wrap**	500 units	**300 units**	**200 units**	**1,000 units**
FIN-402	**Spinach Green Smoothie**	400 units	**400 units**	**0 units**	**800 units**
FIN-403	**Garden Veggie Bowl**	450 units	**150 units**	**0 units**	**600 units**
TOTALS	**3 Finished Batches**	**1,350 (56.2%)**	**850 (35.4%)**	**200 (8.3%)**	**2,400 (100%)**

OUTBOUND SHIPMENTS TELEMETRY (LOTLINE_APP.SHIPMENTS)

Direct manifest queries for finished batches FIN-401, FIN-402, and FIN-403.

Shipment ID	Batch Lot	Product Name	Retail Destination	Quantity	Logistics Status
SHP-001	FIN-401	Green Spinach Wrap	**FreshMart Nagpur Central**	**300 units**	**DELIVERED**
SHP-002	FIN-401	Green Spinach Wrap	**FreshMart Pune West**	**200 units**	**IN_TRANSIT**
SHP-003	FIN-402	Spinach Green Smoothie	**QuickCart Mumbai**	**400 units**	**DELIVERED**
SHP-004	FIN-403	Garden Veggie Bowl	**CityGrocer Nashik**	**150 units**	**DELIVERED**

LOGISTICS INTERVENTION & IN-TRANSIT SHIPMENT MANAGEMENT

INTERVENTION BOUNDARY CLARIFICATION (ACT-002 vs SHP-002):
• **Action Protocol ACT-002** is defined as NOTIFY_RETAILER for FreshMart Nagpur Central (300 units delivered). ACT-002 does NOT constitute proof of carrier contact or transit interception.
• **Active Transit Shipment SHP-002** carries **200.0 units** of Green Spinach Wrap (FIN-401) en route to FreshMart Pune West. Interception is managed via dedicated workflow table LOTLINE_APP.LOGISTICS_INTERVENTIONS (Intervention ID INT-001).
• **Current Intervention State:** Carrier: *ColdChain Logistics Express* | Tracking Ref: *CC-TRK-98442* | Status: **PENDING_APPROVAL**. An interception is reported completed only when a human operator verifies the carrier response.

SECTION 04: RESPONSE ACTIONS & CLOSED-LOOP EXECUTION TRACKING

Operational actions formulated by LangGraph multi-agent workflow and committed to Db2 under human governance.

Action ID	Action Type	Target Entity	Action Description	Status	Approved By
ACT-001	**HOLD_INVENTORY**	LOT: RAW-204	Maintain quarantine of remaining raw-material inventory.	APPROVED	Lead Product Safety Officer
ACT-002	**NOTIFY_RETAILER**	RETAILER: FreshMart Nagpur Central	Prepare retailer notification for potentially affected finished products.	PROPOSED	Pending Review
ACT-003	**TARGETED_RECALL**	LOT: FIN-401	Evaluate affected finished spinach wrap batch for targeted recall.	PROPOSED	Pending Review
ACT-004	**HOLD_DISTRIBUTION**	LOT: FIN-402	Review and prepare a distribution hold for affected smoothie batch.	PROPOSED	Pending Review

CLOSED-LOOP EXECUTION TELEMETRY (COMMUNICATIONS & RECOVERY)

Operational status of consignee communications, product recovery, and dispositions.

Comm ID	Consignee Organization	Implicated Batch	Contact Person	Communication State
COMM-001	**FreshMart Nagpur Central**	FIN-401 (SHP-001)	Rajesh Sharma (Store General Manager)	PENDING_APPROVAL
COMM-002	**FreshMart Pune West**	FIN-401 (SHP-002)	Kavita Deshmukh (Inbound Logistics Manager)	PENDING_APPROVAL
COMM-003	**QuickCart Mumbai**	FIN-402 (SHP-003)	Amitabh Varma (Fulfillment Director)	ACKNOWLEDGED
COMM-004	**CityGrocer Nashik**	FIN-403 (SHP-004)	Sunita Patil (Quality Assurance Lead)	PENDING_APPROVAL

Total Affected Finished Goods	**2,400 UNITS**	**Warehouse Stock Contained**	**1,350 UNITS (56.2%)**
Market Distributed Goods	1,050 UNITS (850 Deliv + 200 Transit)	**Market Recovered (Retail/Transit)**	**0 UNITS (0.0%)**
Outstanding Market Exposure	**1,050 UNITS (Unrecovered)**	**Operational Destruction Verified**	**0 Units | 0.0 KG**

SECTION 05: FSSAI SCHEDULE-I REGULATORY PROFILE & DEADLINE

Field-by-field statutory compliance mapping under the Food Safety and Standards (Food Recall Procedure) Regulations, 2017.

STATUTORY DEADLINE MANDATE (FSSAI 2017 REGULATIONS):
The regulation requires Schedule-I information to be submitted to the competent authority immediately and within **24 hours** of the FBO becoming aware that food requires recall.
• Awareness Timestamp (Db2): **2026-11-09T15:36:48** | Statutory 24-Hr Deadline: **2026-11-10T15:36:48**
• Deadline Evaluation: **EXPIRED** (5.2 hours from discovery alert).

Item	Statutory Field Requirement	Db2 Evidence / Reported Value	FSSAI Verification Status
#1	**FBO Name & Address** Full legal name and registered business/plant address of food business operator.	GreenLeaf Foods, Plant FAC-NGP, Maharashtra, India	Db2 verified
#2	**Contact Person Details** Name, designation, telephone, and email of designated recall coordinator.	Quality Assurance Team (qa@greenleaf.example.in)	Awaiting human verification
#3	**Implicated Food Identity** Product name, brand, mfg date, batch/lot codes, affected quantities, FSSAI licence copy.	Green Spinach Wrap (FIN-401), Spinach Smoothie (FIN-402), Veggie Bowl (FIN-403). Total 2,400 units. Mfg Date: 2026-11-09.	Evidence missing
#4	**Reason for Recall** Detailed biological/chemical reason for recall and risk severity evaluation.	Potential Salmonella contamination detected in raw organic spinach lot RAW-204 propagating into downstream batches.	Db2 verified
#5	**Date & Discovery Circumstances** Date and detailed circumstances under which the hazard/defect was discovered.	2026-11-09T15:36:48.636226 — Supplier QA rapid immunoassay alert during raw intake audit.	Db2 verified
#6	**Nature of Regulatory Violation** Section of FSS Act, 2006 or specific standards violated.	Section 26(1) FSS Act 2006 — Prohibition of Unsafe Food (Microbiological Contaminant: Salmonella). Internal Class I hazard severity.	Db2 verified
#7	**Reporting Person Identity** Name, designation, telephone, and postal address of submitting official.	Reported by: Quality Assurance	Manually required
#8	**Testing Undertaken & Results** Microbiological and chemical testing records, laboratory name, and certificate of analysis.	Intake Screening: Positive for Salmonella spp. Confirmatory reference laboratory plating pending.	Db2 verified
#9	**Total Quantity Manufactured** Total volume/units produced across all implicated finished packaging runs.	2,400 Finished Units (FIN-401: 1,000 | FIN-402: 800 | FIN-403: 600)	Db2 verified
#10	**Duration of Business** Time period over which the implicated batches have been commercialized.	Commercial production run: 2026-11-09. Product commercialization active 8 months.	Manually required
#11	**Quantity Sold into Food Chain** Quantity distributed to trade consignees with documentary bill-of-lading evidence.	1,050 Units Distributed (850 Units Retail Delivered | 200 Units Active In-Transit)	Db2 verified
#12	**Recall Communication & Method** Proposed communications, method of dispatch, and confirmation receipts.	Written formal recall notices drafted for 4 consignees (FreshMart Nagpur, FreshMart Pune, QuickCart Mumbai, CityGrocer Nashik).	Db2 verified
#13	**Actions Taken & Mitigation Plan** Immediate containment steps executed and proposed recovery timeline.	ACT-001 APPROVED (Quarantined 50 KG raw + 1,350 warehouse units). ACT-002..004 PROPOSED. Carrier interception INT-001 initiated for SHP-002.	Db2 verified
#14	**Other Relevant Information** Any other details relevant to food safety, public health, or downstream traceability.	Raw material mass variance: 50.0 KG unexplained intake difference flagged as UNRECONCILED. No consumer illness reported to date.	Db2 verified

SECTION 06: POST-RECALL EFFECTIVENESS & IMMUTABLE AUDIT TRAIL

FSSAI post-recall reporting metrics and chronological audit event trail from LOTLINE_APP.AUDIT_LOG.

Post-Recall Reporting Requirement (FSSAI 2017)	Operational Metric & Evidence Grounding
Original Affected Quantities	2400 Finished Units + 500 KG Raw
Warehouse Stock Contained	1350 Units (56.3% Contained in Facility)
Total Distributed Quantities	1050 Finished Units (850 Delivered + 200 Transit)
Downstream Orgs Identified	4 Commercial Consignees
Consignees Formally Notified	1 Consignees
Consignees Acknowledging	1 Consignees
Market Quantity Recovered	0 Units (0.0% of Market Distributed)
Market Quantity Outstanding	1050 Units in Trade Circulation
Quantity Corrected / Reprocessed	0 Units (Microbial Class I - Reprocessing Prohibited)
Operational Quantity Destroyed	0 Units | 0.0 KG (25.0 KG synthetic test fixtures excluded)
Public Recall Communication	ACT-004 Public Health Warning Prepared
Outstanding Public Health Risks	1050 unrecovered retail units; 50.0 KG raw material mass variance

IMMUTABLE AUDIT TRAIL EVIDENCE (LOTLINE_APP.AUDIT_LOG)

Audit ID	Timestamp	Event Type	Actor	Target	Audit Details
AUD-218	2026-11-10T15:05	**REGULATORY_STAGE_UPDATED**	Compliance Officer	REG-001	FoSCoS regulatory tracker REG-001 moved to stage OPERATOR_VERIFIED. FoSCoS Ref: FOSCO
AUD-217	2026-11-10T15:05	**DISPOSITION_EXECUTED**	Plant Lead	DISP-012	Executed disposition DISP-012 (DESTRUCTION of 5.000 KG for RAW-204). Evidence: TEST-C
AUD-216	2026-11-10T15:05	**DISPOSITION_APPROVED**	Safety Officer	DISP-012	Human operator approved disposition DISP-012 (DESTRUCTION of 5.000 KG for RAW-204). N
AUD-215	2026-11-10T15:05	**DISPOSITION_PROPOSED**	Tester	DISP-012	Proposed disposition DISP-012 for RAW-204 (5.0 KG) as DESTRUCTION. Method: Municipal
AUD-214	2026-11-10T15:05	**LOGISTICS_INTERVENTION_UPDATED**	Logistics Lead	INT-001	Logistics intervention INT-001 for shipment SHP-002 transitioned from CARRIER_ACKNOWL
AUD-213	2026-11-10T15:05	**DOSSIER_GENERATED**	Chief Safety Officer	LOTLINE-RPT-INC-2026-001-150537	Generated internal recall intelligence dossier (SHA-256: abc032a7d4171544...). FSSAI

STATUTORY DISCLAIMER & NON-SIGNATURE NOTICE:

1. **Internal Preparation Document:** This dossier is an internal facility recall intelligence compilation. It does not constitute an official statutory filing with FSSAI or local municipal health authorities. Official notification must be executed by an authorized Food Business Operator officer via the FoSCoS portal.
2. **Cryptographic Integrity Notice:** The running SHA-256 digest embedded across all page footers serves as an automated mathematical proof of document binary integrity and does not constitute a legal digital signature under the Information Technology Act.
3. **Closure Prerequisite Policy:** In accordance with Section 28 of the Food Safety and Standards Act, 2006, a food recall incident cannot be marked CLOSED while unrecovered market units, pending consignee acknowledgements, or unexecuted product dispositions remain unresolved.